

Evolving a hybrid model of a fed-batch fermentation process

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This paper presents a novel method for the system identification of the fed-batch fermentation process defined in the problem statement of the Biotechnological Control Forum Modelling and Control Competition. The identification methodology involves a hybrid of mechanistic modelling and Genetic Programming techniques. It provides an accurate model of the system which should be extremely useful in both the optimisation and control of this process.

Keywords: System modelling; mechanistic modelling; genetic programming; fed-batch fermentation.

Nomenclature

X	Biomass
D	Dilution rate
F_1	Feed rate of substrate 1
F_2	Feed rate of substrate 2
P	Extracellular (measured) product concentration
$V_{d,P}$	Maximum rate of product degradation
$k_{d,P}$	Michaelis-Menten constant for product degradation
$k_{d,X}$	Rate constant for break-up of cells
R_{gen}	Rate of product formation
S_1	Concentration of substrate 1
S_{1f}	Concentration of substrate 1 in feed stream
S_2	Concentration of substrate 2
S_{2f}	Concentration of substrate 2 in feed stream
SGR	Specific growth rate of cells

1. Introduction

In order to encourage research into methodologies suitable for industrial fed-batch process modelling and control, the Biotechnological Control Forum initiated a competition which involved the identification or modelling of a notional fed-batch fermentation process and the design and testing of an optimal controller for the system. The Forum distributed an encrypted computer model that defined the process.

The process in question produced a secondary metabolite (of concentration P) as the product. The micro-organism used two substrates (of concentration S_1 and S_2 , respectively) for growth and product generation. These substrates were supplied to the culture by two

separate feeds (with flowrates F_1 and F_2) of unknown concentration (S_{1f} and S_{2f}). On-line measurements of biomass (X), substrate and product concentrations and the reactor volume (V) were available. The simulation was designed to be a simplified (but accurate) representation of an industrial fed-batch fermenter. As such, the initial conditions and process parameters varied between batches, the fermenter was subject to unmeasured disturbances and noise was added to all measurements. Constraints on the feed flowrates and the maximum volume of the fermenter were also specified ($0 \leq F_1 \leq 50$ units, $0 \leq F_2 \leq 50$ units, $V \leq 4000$ units).

This paper discusses a novel approach to the solution of the first part of this problem, the modelling or system identification of the process.

There are two general methodologies that can be adopted for building a process model. Firstly, a 'bottom-up', mechanistic approach can be employed. This is generally the preferred approach for engineering modelling tasks, but it has the disadvantage of requiring a deep understanding of the basic physical and chemical properties of the system. A common alternative is to approach the problem from the top down and generate an input-output model of the process from plant data only. An example of this approach is a novel technique known as Genetic Programming, which produces relatively simple model structures compared to alternative top-down techniques, such as neural networks (Willis *et al.*, 1990).

In the case of most biochemical systems, the stoichiometry of the important metabolic reactions has been understood for some time. These reaction networks can therefore be incorporated into structured, dynamic models which should be applicable to a wide range of different systems and operating conditions. In previous work (eg, Sanderson *et al.*, 1995), a mechanistic model for such a system has been developed and successfully applied to a range of animal cell cultures.

Unfortunately, such a deep understanding of the system is not always available and this lack of knowledge is particularly true in the case of the simulated fermentation process under consideration here. The nature and concentration of the feeds and products is not known; the organism (or organisms) that are being cultured are not identified and the basic units for the input and output measurements are not supplied (eg, ml/h or gal/s). With such a lack of understanding about the system, data-based modelling might appear to be more appropriate than the mechanistic approach. However, with data-based modelling, algorithms are required to rediscover relationships for which there are already well

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established mechanisms. In addition, the 'black-box' nature of these techniques makes it difficult to build in constraints or observed process characteristics.

For biological systems, previous work (eg, Chotteau and Bastin, 1993) has demonstrated that it is possible to draw on an understanding of the batch fermentation process to provide a mechanistic framework of the process and to use system identification techniques to characterise parts of the system that are poorly understood. This contribution therefore proposes a novel hybrid of mechanistic and Genetic Programming techniques to perform the required system identification task.

2. Genetic programming

Genetic Programming (GP) is a type of evolutionary algorithm. It treats solutions to a given problem as if they were genetic material arranged on a 'chromosome'. A population of these solutions is then used to 'breed' a good solution to the problem through the application of a number of recombination operators that mimic the effect of sexual reproduction. The principle of Darwinian 'survival of the fittest' is used to guide the selection of solutions for reproduction. A flow diagram of a typical GP algorithm is shown in Fig 1.

GP has many similarities to the standard genetic algorithm of Goldberg (1989). However, there are a number of differences between the two paradigms that make GP more suitable for tackling a range of new problem domains. Perhaps the most significant of these is that GP abandons the binary string coding typically adopted for genetic algorithms in favour of a more flexible tree-structured variable-length chromosome (see Fig 2). Koza (1992) presents a comprehensive introduction to the field.

GP can be used for system identification tasks by performing symbolic regression. When used in this manner, the algorithm's recombination operators directly manipulate the structure of a population of mathematical expressions. These operators serve to alter and recombine elements of the expressions in the population. An introduction to the use of GP for modelling process systems and details of the GP algorithm used for this application can be found in McKay *et al* (1996).

The first step in the implementation of GP is the definition of the input (or 'terminal') set. In other words, when developing a mathematical model of a process it is necessary to supply the input variables that are thought to be related in some way to the output. In addition, the algorithm should have the ability to generate constants, as it will generally be a combination of input values and numeric constants that produce the required regression model.

It is also necessary to define a function set. This is the set of arithmetic operations and mathematical functions that the algorithm may use when constructing a potential solution to a given problem. Typically, the function set will include arithmetic operators such as addition and multiplication, common mathematical functions (such as the square root or exponential functions) as well as any specialised functions that may be of some assistance to the algorithm for a particular problem (such as the Michaelis-Menten function commonly used in the modelling of biochemical systems). The search space of the GP algorithm contains

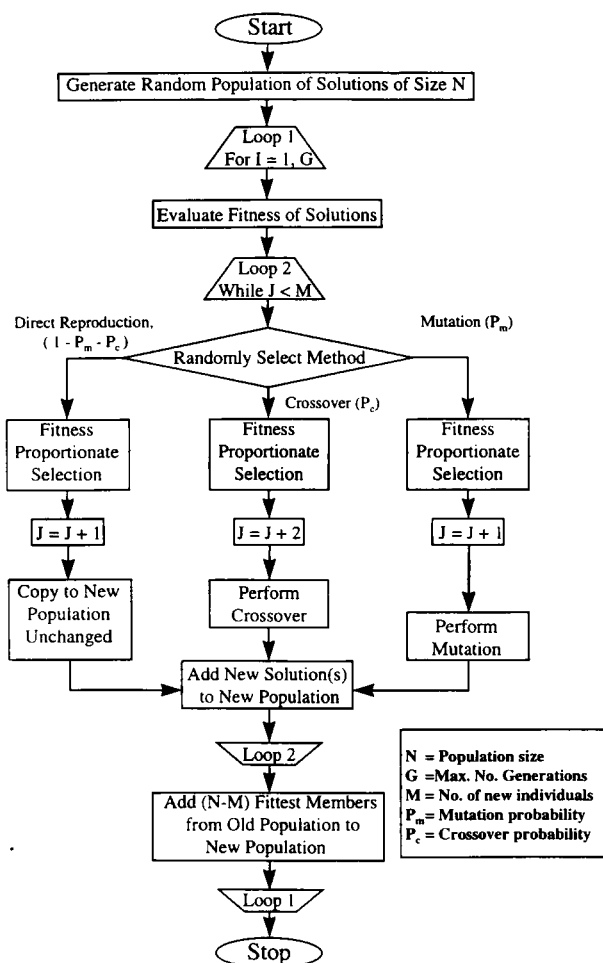


Fig 1 Typical genetic programming algorithm flowsheet

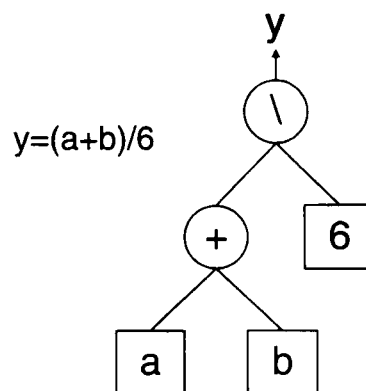


Fig 2 Example of a tree structured chromosome

all possible models that can be constructed from elements in the function and input sets. By applying random variation followed by fitness based selection, the algorithm searches the space of possible mathematical expressions and converges on a functional form that accurately models the available data.

Bettenhausen *et al* (1995) looked at the application of GP to the modelling of fed-batch fermenters. GP was used to evolve dynamic models of the biomass in the fermenter using a function set including locally recursive functions (such as first-order plus dead-time transfer functions and feedforward and feedback loops). While this demonstrated the applicability of GP for the modelling of fed-batch fermenters, computation times

were prohibitively expensive (with a single run taking 24 hours on a cluster of three Sparc 20 and two Sparc 5 workstations). By comparison, run times for the development of static functions using GP are relatively short. This can be utilised in the proposed hybrid modelling approach to reduce model development time, by using the mechanistic framework to capture the dynamics of the system and only requiring GP to identify static relationships.

3. Model development

The first step in developing a hybrid model was to formulate a set of equations that included as much process knowledge as possible and described the five measured variables as functions of time. Of these equations, the simplest was that for reactor volume – in a fed-batch reactor with no overflow, the rate of change of volume is equal to the sum of the feed rates (Eqn (5)).

The specification of an equation describing the rate of change of biomass with respect to time proved more of a challenge, as is shown in Eqn (1). Clearly, the addition of feed streams to the bio-reactor will dilute the concentration of biomass and this term is relatively easy to include. However, during preliminary process analysis it had been noted that the biomass concentration could decrease even when no feed was being supplied – it must, therefore, be subject to some form of degradation. It was assumed that the growth rate was zero when the substrate concentrations were zero and no feed was being supplied, hence the rate of change of biomass was equal to the rate of decomposition. By examining the culture's behaviour, it was observed that the degradation was first-order with respect to biomass and so this feature was included. Finally, the rate of cell growth must be described. No simple correlations (such as that of Monod, 1949) could be found between the substrate concentrations and the rate of growth, so the unknown behaviour was grouped together into a single term for 'specific growth rate' (SGR).

Like biomass, product is diluted by feeding and this effect can easily be described, as seen in Eqn (2). Also like the biomass, the product appears to degrade. At low concentrations (such as those observed under purely batch conditions), this degradation appears to be first-order with respect to the product concentration, while at higher concentrations the degradation appears to be zero-order. Degradation was therefore described by Michaelis-Menten kinetics, which implicitly assume that there is a constant concentration of some degradatory agent in the bio-reactor. No simple term for product formation could be identified, so the production kinetics were grouped together into a single term, R_{gen} . While it is likely that this rate of generation would be proportional to the biomass, such a functionality was not built into the expression because of numerical instability – at low biomass concentrations, measurement inaccuracies had a substantial effect, while at higher concentrations, good predictions would rely on accurate predictions of both the biomass and the rate of formation. It was therefore decided to use the term in its current form. If there were a simple relationship between the biomass and product formation, the GP algorithm should be able to identify

it. The overall rate of change of product concentration is thus described by Eqn (2).

The behaviour of the two substrates is very similar from a modelling point of view and their concentrations are described in Eqns (3) and (4), respectively. In order to characterise the concentrations in the two feed streams, the bio-reactor was run without the addition of feed until the concentration of biomass in the reactor was insignificant ($< 10^{-9}$ units). One feed stream was then turned on for a short period at a high flowrate and the differences in substrate concentration and reactor volume before and after feeding were observed. The rate at which the two substrates are utilised was not found to be related by simple stoichiometric coefficients to the rate of cell growth or product formation, so the rates of use were lumped into single terms $R_{\text{use},S1}$ and $R_{\text{use},S2}$. As was the case with the rate of product formation, the rates of substrate use were not multiplied by the biomass, since this introduced numerical inaccuracies.

$$\frac{dX}{dt} = (SGR - D - k_{d,X})X \quad \dots(1)$$

$$\frac{dP}{dt} = R_{\text{gen}} - V_{d,P} \left(\frac{P}{k_{d,P} + P} \right) - D \cdot P \quad \dots(2)$$

$$\frac{dS_1}{dt} = (S_{1f} - S_1) \frac{F_1}{V} - S_1 \frac{F_2}{V} - R_{\text{use},S1} \quad \dots(3)$$

$$\frac{dS_2}{dt} = (S_{2f} - S_2) \frac{F_2}{V} - S_2 \frac{F_1}{V} - R_{\text{use},S2} \quad \dots(4)$$

$$\frac{dV}{dt} = F_1 + F_2 \quad \dots(5)$$

where the dilution factor is given by,

$$D = \frac{F_1 + F_2}{V} \quad \dots(6)$$

If it is assumed that static relationships exist between the available system variables (that is, biomass, product and substrate concentrations, volume, time, and the feedrates) and the unknown rates in Eqns (1)–(4) (that is, SGR , R_{gen} , $R_{\text{use},S1}$ and $R_{\text{use},S2}$), then a static GP algorithm can be used to identify a relationship for each of the rates.

Eqns (1) to (4) may be used to solve directly for SGR , R_{gen} , $R_{\text{use},S1}$ and $R_{\text{use},S2}$ if reliable estimates of the derivatives are available. Such estimates can be obtained from plant data, as long as some care is taken to select an appropriate sampling period and to minimise the effect of process noise. A sampling period of 0.25 time units was found to be sufficient for capturing the fastest dynamic responses observed. A second-order Butterworth filter (Oppenheim and Shaffer, 1975) was then used to reduce process noise and smooth any discontinuities.

4. Implementation aspects

In order to provide training data, 20 fed-batch experimental data sets, each covering 200 time units, were generated. A further ten data sets were produced for use in model verification. The feed flowrates for each batch were generated as multi-level pseudo-random signals, subject to the feed flowrate and fermenter volume constraints specified in the problem statement.

The GP algorithm used has been implemented in MATLAB on a Pentium-90 PC and typical run times of the order of 100 min were achieved. However, the underlying MATLAB environment is particularly inefficient at performing certain functions, so considerable further improvements could be made if key parts of the algorithm were coded in C or FORTRAN (and compiled as MEX files).

The basic input set supplied to the GP algorithm included all of the measured process variables (X, S_1, S_2, P, V), along with the two feed flowrates (F_1 and F_2) and the time since the start of a batch (T). The basic function set used by the GP algorithm included the mathematical operators $+$, $-$, $/$, $*$, as well as the mathematical functions exponential, natural logarithm, square root, square and power. During initial experiments, it was found that the GP algorithm was capable of producing a reasonable approximation to the majority of the data. However, further improvements were obtained by supplying the GP algorithm with two functions that are commonly used in biological modelling: the Michaelis-Menten equation and an inverse Michaelis-Menten function. These functions are described by Eqns (7) and (8), respectively,

$$y = V_m \frac{k}{k + x} \quad \dots (7)$$

$$y = V_m \frac{x}{k + x} \quad \dots (8)$$

where V_m and k are constants.

Further numerical experiments using the GP algorithm provided reasonable expressions for the unknown rate terms. However, it was thought that the quality of model approximation and robustness could be improved still further. It is in addressing these problems that the GP algorithm provides significant benefits. For example, one of the features of the model's mechanistic framework is that there was no mass balance that linked substrate utilisation to the rate of cell growth or product formation. This led to undesired model behaviour. Under certain circumstances, the substrate concentrations would approach a constant non-zero level while the biomass continued to increase rapidly. Clearly, this behaviour is extremely unlikely to persist in reality. In order to rectify this, the GP algorithm was modified in that it was required to add the term $\alpha.SGR.X$ (where α was a modelling constant) to all members of the population when evolving expressions for $R_{use,S1}$ and $R_{use,S2}$. Incorporating this process knowledge eliminated unrealistic model behaviour, and more importantly improved the stability of the resultant model. It was also found that product formation usually only occurred while the cells were growing. The GP model for R_{gen} could therefore be improved by including SGR in its input set.

The final set of equations that were evolved by the GP algorithm are summarised by Eqns (9)–(12). It should be noted that the ability to provide user-defined functions (Eqns (7) and (8)) and partial structure to the GP algorithm are key strengths of this work that have substantially strengthened the model's predictive ability.

$$SGR = f(T, F_1, F_2 X, S_1, S_2, P, V) \quad \dots (9)$$

$$R_{gen} = f(T, F_1, F_2 X, S_1, S_2, P, V, SGR) \quad (10)$$

$$R_{use,S1} = f(T, F_1, F_2 X, S_1, S_2, P, V) + \alpha_1 SGR X \dots (11)$$

$$R_{use,S2} = f(T, F_1, F_2 X, S_1, S_2, P, V) + \alpha_2 SGR X \dots (12)$$

The validity of any modelling paradigm is limited to the operating region for which the model has been developed. One of the difficulties with input–output models is that their behaviour can become unpredictable when asked to extrapolate beyond the region of their training data. However, the hybrid model's mechanistic framework made it a simple matter to explicitly include bounds and constraints. Soft constraints¹ were introduced to bound the values of SGR , R_{gen} , $R_{use,S1}$ and $R_{use,S2}$ when the process moved into a region outside the range of the training data. Soft constraints were also used to prevent the model from entering regions that are physically meaningless. For example, it is clearly not possible for the concentration S_1 to become negative. A soft constraint can therefore be used to force the GP prediction for $R_{use,S1}$ to zero as S_1 approaches this lower bound. The model is thereby prevented from trying to enter an unrealistic region. These measures improved the overall stability of the model and made its extrapolative behaviour more predictable.

In order to integrate the set of differential and algebraic equations that describe our model, both SPEEDUP and the numerical integration algorithms in the MATLAB ODE suite were used. SPEEDUP proved to be faster and more robust with better diagnostics than the best available MATLAB function, and so it was used in the developmental stage of the project. However, due to the flexible nature of the MATLAB environment, it was a more convenient choice for implementing the model as an ' n -step ahead predictor'. As a result, MATLAB was used to generate the results presented in the next section.

5. Results

The expressions obtained by the GP algorithm for the rates are given below. The bar above a variable name indicates that the variable has been scaled prior to being presented to the GP algorithm. Note the appearance of the Michaelis-Menten type expression and the $SGR.X$ term in Eqn (13).

$$\begin{aligned} \bar{R}_{gen} = & -0.276\bar{X} - 0.137\bar{T} + 0.119\bar{X}\bar{F}_1 - 1.09\bar{F}_2 \\ & + \frac{\bar{S}_2}{1.26 \times 10^{-5} + \bar{S}_2} \\ & \times (0.163 + 0.2\bar{X} - 3.56\bar{S}GR\bar{P} + F_2) \\ & + \bar{S}GR\bar{X} + 4.54\bar{S}GR + 0.154 \quad \dots (13) \end{aligned}$$

$$\begin{aligned} \bar{S}GR = & -0.112 - 0.0103\bar{S}_2 + 0.192\bar{F}_2 - 1.98\bar{X}\bar{S}_2 \\ & - 0.0773\bar{X}\bar{F}_2 - 0.542\bar{X} + 0.268\bar{X}^2 \\ & + \bar{X}\bar{S}_2^2 - 0.0981\bar{X}\bar{V} + 0.123\bar{S}_1 \\ & + (2.13e^{-5})\bar{T} \quad \dots (14) \end{aligned}$$

¹Implemented as sigmoid functions.

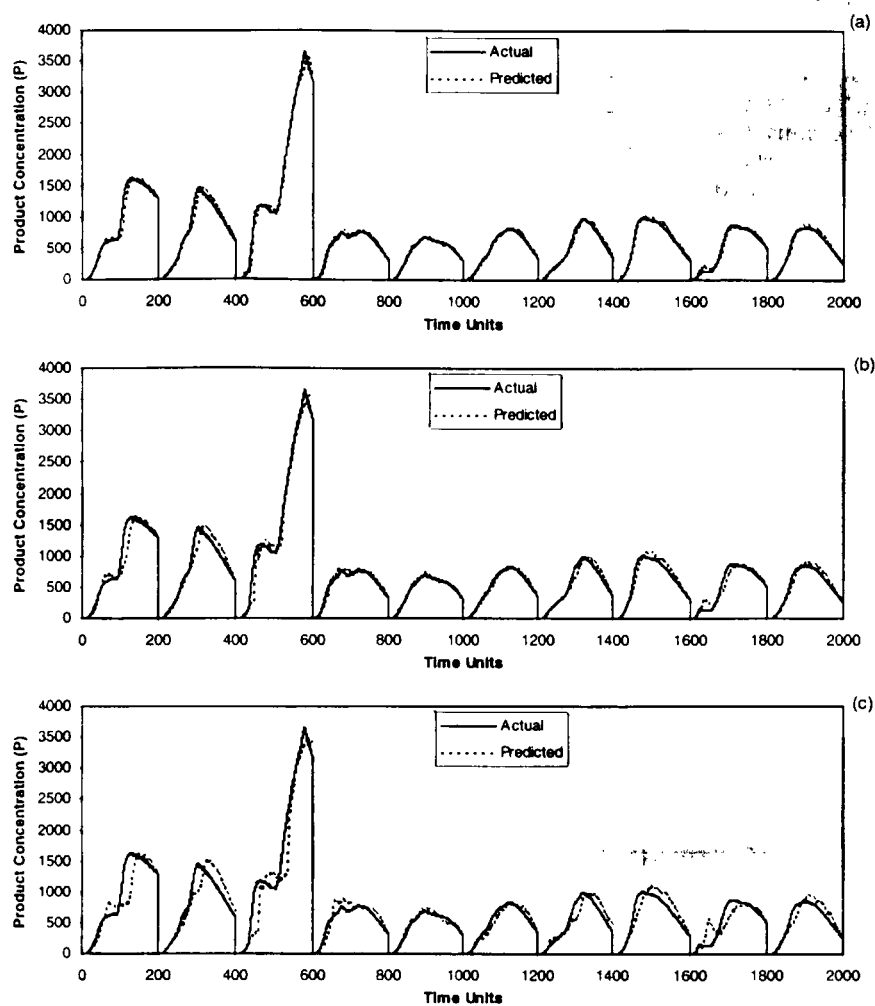


Fig 3 Model performance predicting product concentration (P) n -steps ahead: (a) $n = 25$, (b) $n = 50$, (c) $n = 100$

$$\begin{aligned} \bar{R}_{\text{use}, S_2} = & 9.48 \times 10^{-2} \bar{S}GR\bar{X} - 0.837\bar{T} + 0.118\bar{F}_1 \\ & + 6.51 \times 10^{-2}\bar{F}_2 + 1.14\bar{X} + 0.138\bar{S}_1^3 \\ & + 1.13 \times 10^{-2}\bar{S}_1 + 0.355\bar{S}_1\bar{T} - 0.179\bar{S}_1\bar{F}_1 \\ & + 2.40 \times 10^{-2}\bar{S}_1\bar{F}_2 - 1.39\bar{S}_1^2\bar{X} \\ & + 1.33\bar{S}_1^2\bar{T} - 0.156 \end{aligned} \quad \dots(15)$$

$$\begin{aligned} \bar{R}_{\text{use}, S_2} = & 2.64 \times 10^{-4} \bar{S}GR\bar{X} - 9.45 \times 10^{-2} \\ & - 1.81e^{-2}\bar{S}_1\bar{F}_1 + 0.246\bar{F}_2\bar{S}_2^2 \\ & + 0.470\bar{S}_2\bar{F}_2 + 0.428\bar{F}_2 + 3.65 \times 10^{-2}\bar{V}\bar{F}_2 \\ & + 0.327\bar{S}_2\bar{V}\bar{F}_2 - 0.248\bar{S}_2\bar{V}^2\bar{F}_2 \\ & - 0.644\bar{S}_2^2 - 0.226\bar{S}_2\bar{V}^2 - 0.220\bar{V} \end{aligned} \quad \dots(16)$$

The performance of the model as a 25-, 50- and 100-step-ahead predictor of both P and X on the 10 batches of the verification data set is demonstrated in Figs 3 and 5, respectively. The results for S_1 , S_2 and V are of similar accuracy (V is almost perfect as it only involves integrating the input flowrates once the difference in units between the flowrates and volume has been accounted for), but are not presented due to space limitations.

Figs 4 and 6 show the Root Mean Square (RMS)² error between the predicted and actual values of P and X on the verification data increasing linearly with the prediction horizon (ie, the number of 0.25 time units steps predicted ahead). It is worth noting that the model performs exceedingly well for a prediction horizon of at least 100 steps, and remains stable at a prediction horizon of 800 steps (ie, a complete 200 time unit batch length) for all ten batches in the verification data set. The performance of the model when predicting P and X for the whole batch (given accurate initial conditions) is demonstrated in Figs 7 and 8. Clearly there are some model inaccuracies, but the effect of variations in the feed concentration will also be particularly pronounced over these longer horizons.

It can clearly be seen that the model is very accurate over a range of operating conditions and that it could now be used in a number of potential applications. The differences between the model and the experimental data have their origins in a number of places, notably model inaccuracy and unmeasured disturbances such as variation in the concentration of substrates in the feed streams. Little can be done about the former problem, but if this were a real plant it does not seem unreasonable that the concentration of the substrates in the feed

² $RMS = \sqrt{\{\sum_{i=1}^N (y_i - y_{p,i})^2 / N\}}$ where y and y_p are the actual and predicted output, and N is the number of points.

Fig 4 Model performance predicting product concentration (P) as a function of the prediction horizon

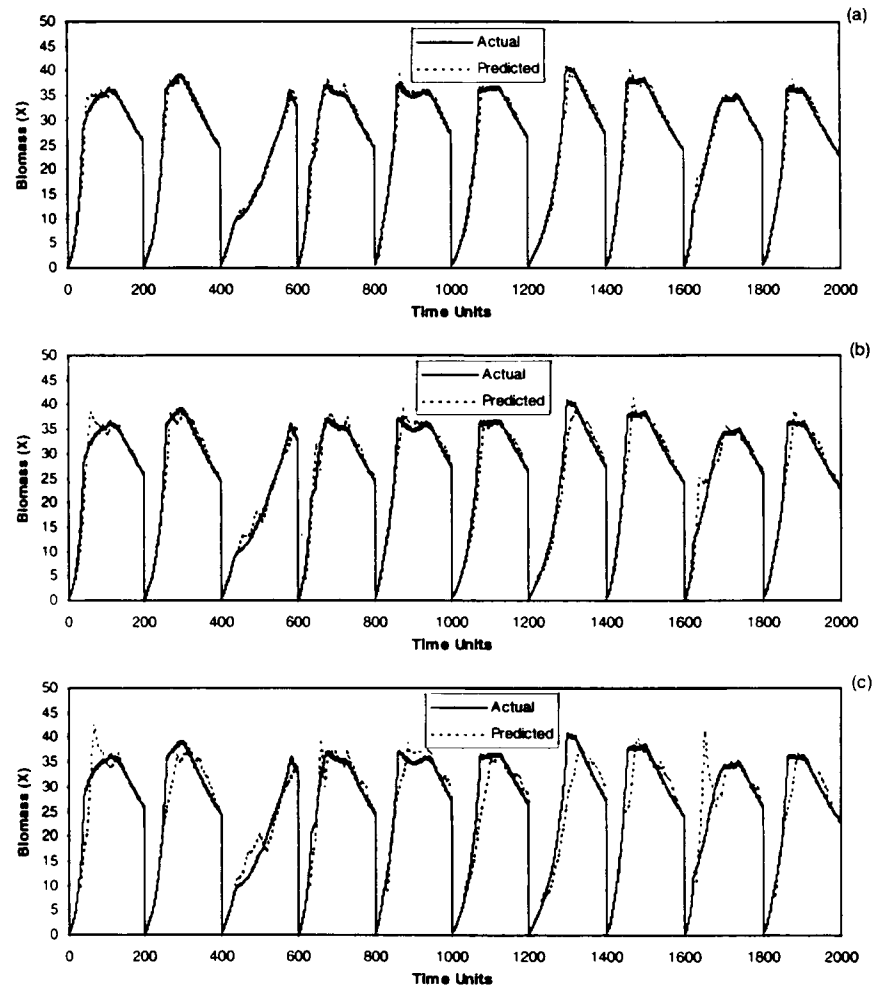
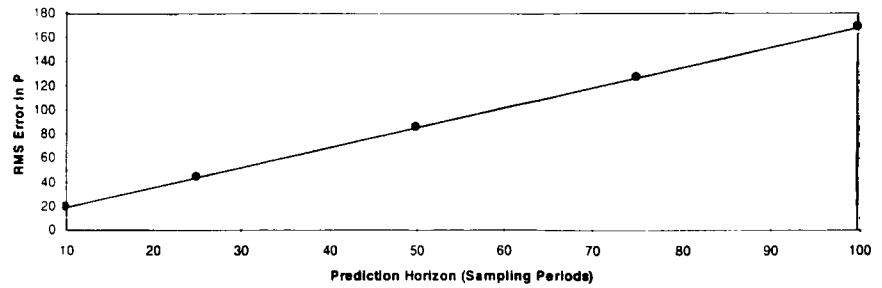


Fig 5 Model performance predicting biomass (X) n -steps ahead: (a) $n = 25$, (b) $n = 50$, (c) $n = 100$

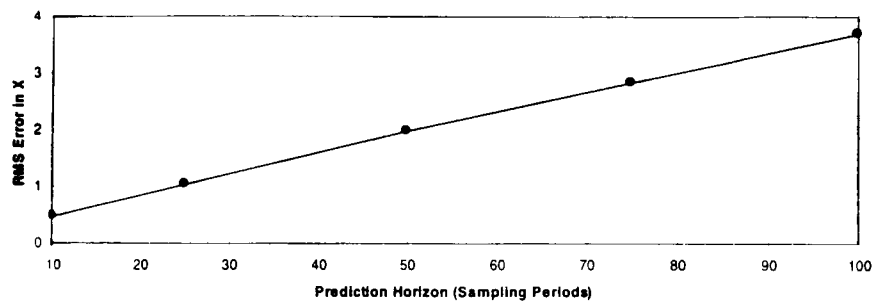


Fig 6 Model performance predicting biomass (X) as a function of the prediction horizon

streams would be measured. Clearly, the technology to measure these concentrations exists, since there are probes in the fermenter. The substrates do not appear to degrade, and so (if feed disturbances were infrequent)

the measurements could even be performed off-line in a laboratory. Measurements of feed disturbances would greatly improve the performance of the model as a long-term predictor.

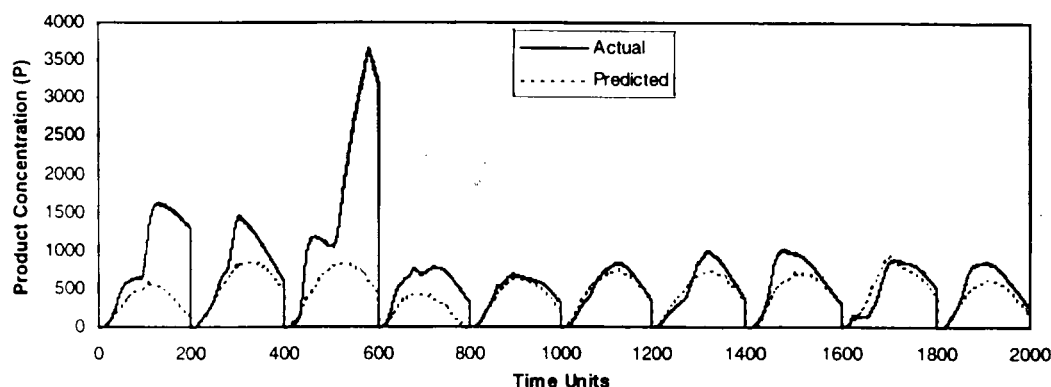


Fig 7 Model performance as a long-term predictor of product (P) for 10 successive (200 time unit) batches

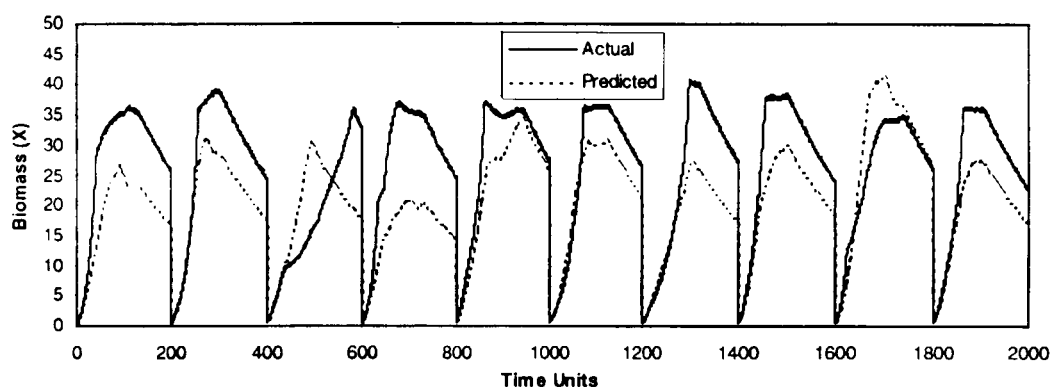


Fig 8 Model performance as a long-term predictor of biomass (X) for 10 successive batches

6. Conclusions

The predictions of the overall model that has been produced by this hybrid modelling technique are accurate, especially when used as an n -step-ahead predictor. Under these conditions, the effects of variations in feed concentration can be minimised. The model is relatively compact and can be solved rapidly, so it should be particularly useful in the areas of optimisation and control.

The use of a hybrid of structured modelling and GP gives access to the best features of the two techniques and provides a model that is likely to perform well under circumstances where neither technique is likely to succeed on its own. GP provides an excellent basis for combining input-output modelling with a partial understanding of a system, ensuring that obvious relationships do not need to be rediscovered.

The modelling technique developed here is by no means specific to bio-reactor simulation; it could be used for any system identification problem in which there is a partial understanding of the system's behaviour. One particularly interesting area for future research will be the reduction of large existing models, such as the animal cell culture model of Sanderson *et al* (1995) – the smaller hybrid model would be expected to run significantly more quickly and to be of use in a variety of applications.

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